Internship Report

Company's name: Agence Spatial Algérienne.
Place of internship: Centre d'exploitation des systèmes de télécommunications spatial.
Name and Position: BENDIF BESMALA , stagiaire.
Internship Period: from 13/09/2024 to 08/10/2024 .
University Name: National Higher School Of Artificial Intelligence.

I would like to thank my supervisors, Sir Mohammad Amin Larabi , Sir Meziane Iftene and Mr Radhwan, for their guidance and support throughout my internship at ASAL . Special thanks to my mentors for sharing valuable insights and helping me build essential skills. I am also grateful to my team members for their collaboration and encouragement, which made this experience both enjoyable and rewarding.

Introduction

Overview of the company:

The Algerian Space Agency (ASAL) is a national public institution with legal and financial independence, established by Presidential Decree No. 02-48 on January 16, 2002, under the Prime Minister's authority.

ASAL is responsible for developing and implementing Algeria's national space policy, aiming to make space technology a driver of the country's economic, social, and cultural development while promoting national security and well-being.

Key roles include:

1. Advising the government on national space strategy and overseeing its execution;
2. Building space infrastructure to strengthen national capacities;
3. Managing and assessing national space activity programs;
4. Proposing space systems aligned with national priorities and overseeing their development;
5. Recommending policies for bilateral and multilateral cooperation;
6. Monitoring Algeria's commitments in regional and international space agreements.

Internship Context and Objectives

During my internship at the Algerian Space Agency (ASAL), I was part of the ASAL team, we focused on many projects related to remote sensing technology that utilise satellites.

The primary objectives of my internship were to:

1. Gain hands-on experience in processing and analysing remote sensing data.
2. Contribute to projects aimed at monitoring environmental changes and supporting urban planning through satellite imagery.
3. Develop a deeper understanding of remote sensing applications and workflows used at a national space agency.
4. Understand the real applications of artificial intelligence in remote sensing fields using real world data.
5. Develop my skills in AI and have real experience with working on real and more complicated projects.

Through this internship, I aimed to strengthen my technical skills in remote sensing, apply analytical techniques to real-world data, and gain insights into the operational and strategic goals of ASAL in advancing Algeria's space initiatives.

Tasks, Achievements, and Learning Outcomes

Tasks:

During my internship at the Algerian Space Agency (ASAL), I was primarily responsible for performing many projects within groups , the tasks we have done included:

- In the first week , we had theoretical courses on general image processingI , AI for advanced image processing , GeoAI jobs , introduction to remote sensing, data augmentation techniques , machine learning and deep learning. We learned many different techniques that we can apply in remote sensing, including also some tps and mini exercises.

- We had an everyday online Q & A session during the internship month , where we answered many different questions on several topics of remote sensing, and we presented them to our supervisors.

- In addition to daily doses on AI , which were covering various topics of AI and application in satellite imagery and remote sensing , they gave us a great amount of valuable information.

- In second week , it was a week full of mini projects , they included :

  1."Transferring Deep Convolutional Neural Networks for the Scene Classification of High-Resolution Remote Sensing Imagery":Reproduce the given scientific paper approach, and enhance the results using any technique you learned in our training course. We used vgg and Resnet models .

2. Using Transformers in Remote Sensing Scene Classification , based on a research paper or developing our own method with a small web application that uses the optimised model, loads an image and shows its class.

3. Remote Sensing Semantic Classification and more precisely, the Binary Semantic classification.(worked on building detection , road extraction , cloud detection , and cloud shadow detection , we used DeepLabv3+ and UNet models).

4. FLOODNET Track 1 (Semi-supervised Classification and Semantic Segmentation on floonet dataset , we worked on multiclass classification).

5.Multiclass Remote Sensing Semantic Classification with Deep Learning:

A. Created shapefiles from Google Earth Engine for various land cover classes and trained a machine learning model in Python using extracted satellite imagery features, achieving accurate classification and insightful visualisations of the results.

B. Create a shapefile using Meringue mobile app and train ML models.

C. Use the ESA WorldCover product for AI model training and application.

6. multiclass semantic classification that uses a raster image and its corresponding ground truth as a VECTOR data.

Task1:

Apply EDA

Enhance the Notebook quality by refining the code and enrich it with results visualisation

Compare results with other classifiers XgBoost, LightGBM; CATBoost; 1D-CNN, SVM

Explain Features importance for each model. E.g: [SHAP, LIME for deep model]

Plot results for qualitative comparison

Task2:

Instead of using the given training GeoTiff file [Trans_nzoia_2019_05-02.tif]; you should use the vector file to download the corresponding Sentinel-2 Tiff image from GEE and then perform the same processing similarly as Task1.

Task3:

Repeat the same using LANDSAT images from GEE

7. SuperResolution Pansharpening HyperSharpening course : with three mini projects which are : Adapt pix2pix for Pan-Sharpening /Hyper-Sharpening , Select and try at least 3 deep learning methods on a Benchmark (dataset).

8. HyperSpectral Images Classification: Implement 3 different state-of-the-art AI-based hyperspectral image classification techniques on any Benchmark.

9. **Multi-scale Object Detection in High-Resolution Satellite Imagery:**Develop a deep learning model for detecting various objects in high-resolution satellite imagery across multiple scales. Address challenges of varying object sizes, dense distributions, and complex backgrounds. Implement and compare different object detection architectures (e.g., Faster R-CNN, YOLO, RetinaNet) adapted for satellite imagery. Evaluate performance on different object classes and scales.

10. **Supervised Object Counting in Satellite Imagery:** Develop a supervised deep learning model for accurate object counting in satellite imagery using fully annotated datasets. Explore and compare different architectures including CNN-based approaches, detection-based counting methods, and regression-based density estimation techniques. Address challenges such as varying object densities, image resolution, and environmental factors. Investigate multi-task learning approaches combining counting with localization or classification.

11. **Weakly Supervised Object Counting in Satellite Imagery:** Create a weakly supervised learning approach for counting objects in satellite imagery using only image-level count annotations. Explore density estimation techniques and attention mechanisms for accurate counting of objects like cars, ships, or solar panels. Implement and compare different weakly supervised counting methods. Develop strategies to handle varying object densities and sizes.

12. **Foundation Model-based Object Detection and Instance Segmentation in Satellite Imagery:** Leverage a pre-trained foundation model (e.g., CLIP, ViT) for object detection and instance segmentation in satellite imagery. Fine-tune the model on satellite imagery datasets. Explore techniques such as prompt engineering, few-shot learning, and zero-shot transfer. Compare performance and efficiency with traditional models trained from scratch.

13. **Foundation Model-based Semantic Segmentation in Satellite Imagery:** Create a versatile segmentation model for satellite imagery inspired by the Segment Anything Model (SAM). Adapt a large-scale foundation model for semantic segmentation tasks in satellite imagery. Implement prompt-based segmentation for flexible and dynamic segmentation based on text or point prompts. Explore few-shot and zero-shot learning capabilities to segment new or rare classes. Develop strategies to handle unique challenges of satellite imagery and create an interactive interface for user-guided segmentation.

- We had a session of Introduction to Google Earth Engine with application to environment monitoring with the guest Speaker: Dr. Achraf Djerida from the Algerian Space Agency. This session promised to be an opportunity to learn about cutting-edge techniques in remote sensing and water resource management. Dr. Djerida will share insights on:

• The importance of environmental monitoring in today's world.

• How Google Earth Engine is revolutionising environmental monitoring.

• Practical applications of satellite imagery.

• Real-world case studies from Algeria and beyond.

- We had also a earning opportunity: "Diving Deep into SAR Images Analysis":

  • The game-changing potential of SAR technology

  • Cutting-edge analysis techniques you need to know

  • Real-world applications that are shaping our understanding of the Earth

  • Hands-on examples that bring theory to life

- "From Air Quality to Methane Emissions: Unravelling Earth's Atmospheric Mysteries" with WIdad Belkadhi , where we discovered:

  • The ABCs of Air Quality: Demystifying pollution and its impacts.

  •Methane Unveiled: The powerful greenhouse gas you need to know about.

  • Where's it coming from? Explore the sources and sectors behind methane emissions.

  • Cutting-edge Science: How we're detecting and measuring methane in our atmosphere.

- presentation for the course 'Introduction to SAR and Its Applications.' Please review it to improve your understanding of the key concepts and practical uses of SAR with Benali Sara.

- The last week was for hackathon series , we collaborated with team members to complete two hackathons , they was about :

  Hackathon 1 was about Remote Sensing Based Algerian Forest Mapping including two tracks , first track was on Algerian Forest Mapping with Noisy Labels and Few Clean Data , in this hackathon track we developed robust forest mapping models for Algeria using a combination of noisy labels from the ESA World Cover map and a small set of manually created clean data. The primary goal is to leverage the large-scale but potentially noisy ESA data while effectively integrating the limited clean labels to improve model performance and generalisation. we addressed key challenges such as label noise, domain adaptation between the ESA data and local Algerian forests, and efficient utilisation of the small clean dataset. The track encourages the exploration of advanced techniques including semi-supervised learning, noise-robust loss functions, transfer learning, etc. to create accurate and reliable forest maps of Algeria.

Second Track :Algerian Forest Mapping with Noisy Labels from ESA World Cover. This hackathon track challenges to develop advanced forest mapping models for Algeria using exclusively the noisy labels provided by the ESA WorldCover map, without any external clean data. The primary objective is to design robust algorithms capable of learning effectively from potentially noisy and mislabeled data across the entire country. We addressed key challenges such as dealing with label noise, mitigating potential biases in the ESA data, and ensuring model generalization across diverse Algerian forest ecosystems. The track encourages the exploration of cutting-edge approaches including noise-robust loss functions, self-supervised pre-training, and ensemble methods to improve model performance in this challenging scenario of learning from noisy labels at a national scale.

Third Track :Algerian Forest Mapping using Transfer Learning from European Forests. This hackathon track challenges to develop innovative forest mapping models for Algeria using transfer learning techniques applied to data from European forests. Leveraging only the ESA WorldCover map data over Europe, without any data from North Africa, we created models that can effectively adapt to the different ecosystems found in Algeria. The primary objective is to explore and implement advanced transfer learning and domain adaptation techniques that can bridge the gap between European and Algerian forest characteristics. Key challenges include addressing the domain shift between these distinct geographic regions, handling differences in forest types and structures, and ensuring model generalization despite the lack of target domain data during training. We explored cutting-edge transfer learning methods, unsupervised domain adaptation techniques, and ways to incorporate prior knowledge about the differences between European and North African forest ecosystems.

- Last thing we did is get a simple presentation on the methodology of writing a research pr scientific paper. It was presented by Mr Meziane , and then I wrote my own research paper about the methodology I did in the second hackathon.

Achievements:

Throughout the internship, I was able to:

Successfully answered all the Q&A sessions and completed all the assigned projects and hackathons .

What I Learned:

This internship allowed me to deepen my understanding of remote sensing and its practical applications. I gained proficiency in data processing software, and developed analytical skills in image classification and feature extraction , also I got an overview on how to use the latest techniques of AI like transformers , fine tuning ,foundation models and lots more. Working closely with experts at ASAL also taught me the importance of precision and collaboration when handling data that can influence policy and planning.I had the opportunity to learn how to manage the work in teams , and i understood how i should learn and export and search and develop something useful in small amount of time , i learned to to handle the pressure,

Overall, my time at ASAL provided me with valuable technical skills, a clearer understanding of remote sensing's impact on national development, and a solid foundation for pursuing a career in the field.